Hilbert Projection Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A minimizing sequence need not converge merely because its distances converge. The parallelogram identity converts near-minimality into the missing Cauchy estimate. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

FA-02 / Functional Analysis and Measure Theory / Report v1

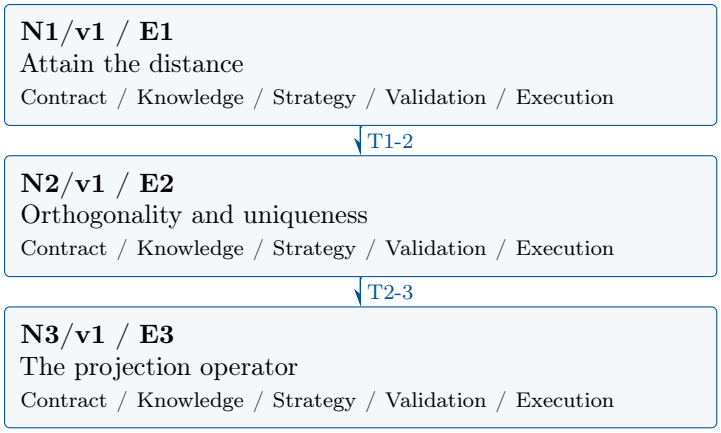
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let H be a real or complex Hilbert space and $M \subset H$ a closed linear subspace. For each $x \in H$ there exists a unique $m \in M$ minimizing $\|x - m\|$. Moreover $x - m \perp M$, and the map $P_M x = m$ is linear with $\|P_M x\| \leq \|x\|$. Inner products are linear in the first variable.*

A. Foundations and permitted imports

Use completeness of H, the parallelogram identity, Cauchy–Schwarz, and the polarization rules for a complex inner product. A closed subspace contains limits of its convergent sequences. The infimum property supplies a sequence approaching a finite infimum.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There is $m \in M$ with $\|x - m\| = d$.

Receiving activity. N2.1

Compatibility. E1/v1, a minimizer m, and each $v \in M$. Check: In a complex space, real variations alone only eliminate the real part of the inner product.

T2-3: N2/E2/v1 to N3. $x - m \perp M$, and this condition characterizes the unique minimizer.

Receiving activity. N3.1

Compatibility. E2/v1 for every $x \in H$. Check: The argument uses that M is a linear subspace, not merely a closed set.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Attain the distance

Object and role. Use the midpoint in the subspace to control differences of two minimizing terms.
Local obligation: There is $m \in M$ with $\|x - m\| = d$.

Inputs. $d = \inf_{m \in M} \|x - m\|$ and a sequence $m_n \in M$ with $\|x - m_n\| \to d$.

C — Contract

Publish E1/v1 only after establishing this obligation: There is $m \in M$ with $\|x - m\| = d$.

K — Knowledge

$d = \inf_{m \in M} \|x - m\|$ and a sequence $m_n \in M$ with $\|x - m_n\| \to d$. Retain the stated assumptions and import versions.

S — Strategy

Use the midpoint in the subspace to control differences of two minimizing terms.

V — Validation

The midpoint lies in M; completeness gives a limit in H and closedness places it in M.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Since $0 \in M$, $0 \le d \le \|x\|$. Choose m_n with $\|x - m_n\| < d + 1/n$. The parallelogram identity gives $\|m_n - m_k\|^2 = 2\|x - m_n\|^2 + 2\|x - m_k\|^2 - 4\|x - (m_n + m_k)/2\|^2$. The final squared norm is at least d^2.

N1.2 The resulting upper bound tends to zero as $n, k \to \infty$. Thus (m_n) is Cauchy and converges in H to some m. Closedness gives $m \in M$, and continuity of the norm gives $\|x - m\| = d$.

Export E1/v1. There is $m \in M$ with $\|x - m\| = d$.

Acceptance gate. The midpoint lies in M; completeness gives a limit in H and closedness places it in M.

Targeted repair. If the minimizing sequence is declared convergent without an estimate, insert the parallelogram bound; if M is not closed, the original attainment claim needs a changed hypothesis. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Orthogonality and uniqueness

Object and role. Vary the minimizer along real and imaginary scalar multiples of v. **Local obligation:** $x - m \perp M$, and this condition characterizes the unique minimizer.

Inputs. E1/v1, a minimizer m, and each $v \in M$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $x - m \perp M$, and this condition characterizes the unique minimizer.

K — Knowledge

E1/v1, a minimizer m, and each $v \in M$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Vary the minimizer along real and imaginary scalar multiples of v.

V — Validation

In a complex space, real variations alone only eliminate the real part of the inner product.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Write $r = x - m$. For real t, minimality implies $0 \le \|r - tv\|^2 - \|r\|^2 = -2t\,\mathrm{Re}\langle r, v\rangle + t^2\|v\|^2$. Taking small positive and negative t forces $\mathrm{Re}\langle r, v\rangle = 0$. In the complex case repeat with iv to get the imaginary part zero.

N2.2 For any $w \in M$, orthogonality gives $\|x - w\|^2 = \|r\|^2 + \|m - w\|^2$. Hence equality with the minimum holds only for $w = m$. Conversely an orthogonal residual gives this identity and therefore a minimizer.

Export E2/v1. $x - m \perp M$, and this condition characterizes the unique minimizer.

Acceptance gate. In a complex space, real variations alone only eliminate the real part of the inner product.

Targeted repair. If only real orthogonality has been checked, add the iv variation and revalidate uniqueness. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: The projection operator

Object and role. Apply uniqueness of the orthogonal decomposition to sums and scalar multiples.
Local obligation: P_M is linear and contractive: $\|P_M x\| \leq \|x\|$.

Inputs. E2/v1 for every $x \in H$. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: P_M is linear and contractive: $\|P_M x\| \leq \|x\|$.

K — Knowledge

E2/v1 for every $x \in H$. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Apply uniqueness of the orthogonal decomposition to sums and scalar multiples.

V — Validation

The argument uses that M is a linear subspace, not merely a closed set.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For $x, y \in H$ and scalars a, b, $aP_M x + bP_M y$ lies in M, while $a(x - P_M x) + b(y - P_M y)$ is orthogonal to M. E2 identifies this as the unique decomposition of $ax + by$, proving linearity.

N3.2 The decomposition $x = P_M x + (x - P_M x)$ is orthogonal, so $\|x\|^2 = \|P_M x\|^2 + \|x - P_M x\|^2$. This proves the norm bound and completes all assertions.

Export E3/v1. P_M is linear and contractive: $\|P_M x\| \leq \|x\|$.

Acceptance gate. The argument uses that M is a linear subspace, not merely a closed set.

Targeted repair. If projection onto a nonlinear closed set is claimed linear, restore the subspace contract; linearity is not part of the general convex projection theorem. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Attain the distance

1. **N1.1.** Since $0 \in M$, $0 \le d \le \|x\|$. Choose m_n with $\|x - m_n\| < d + 1/n$. The parallelogram identity gives $\|m_n - m_k\|^2 = 2\|x - m_n\|^2 + 2\|x - m_k\|^2 - 4\|x - (m_n + m_k)/2\|^2$. The final squared norm is at least d^2.

2. **N1.2.** The resulting upper bound tends to zero as $n, k \to \infty$. Thus (m_n) is Cauchy and converges in H to some m. Closedness gives $m \in M$, and continuity of the norm gives $\|x - m\| = d$.

B. N2 — Orthogonality and uniqueness

1. **N2.1.** Write $r = x - m$. For real t, minimality implies $0 \le \|r - tv\|^2 - \|r\|^2 = -2t\,\mathrm{Re}\langle r, v\rangle + t^2\|v\|^2$. Taking small positive and negative t forces $\mathrm{Re}\langle r, v\rangle = 0$. In the complex case repeat with iv to get the imaginary part zero.

2. **N2.2.** For any $w \in M$, orthogonality gives $\|x - w\|^2 = \|r\|^2 + \|m - w\|^2$. Hence equality with the minimum holds only for $w = m$. Conversely an orthogonal residual gives this identity and therefore a minimizer.

C. N3 — The projection operator

1. **N3.1.** For $x, y \in H$ and scalars a, b, $aP_Mx + bP_My$ lies in M, while $a(x - P_Mx) + b(y - P_My)$ is orthogonal to M. E2 identifies this as the unique decomposition of $ax + by$, proving linearity.

2. **N3.2.** The decomposition $x = P_Mx + (x - P_Mx)$ is orthogonal, so $\|x\|^2 = \|P_Mx\|^2 + \|x - P_Mx\|^2$. This proves the norm bound and completes all assertions.

□

V. WORKED ORCHESTRATION

In $\mathbb{R}^3$, let $M = \{(a, b, 0) : a, b \in \mathbb{R}\}$ and $x = (2, -1, 3)$. The candidate $m = (2, -1, 0)$ has residual $(0, 0, 3)$ perpendicular to M. For $(a, b, 0)$ the squared distance is $9 + (2-a)^2 + (-1-b)^2$, certifying the unique minimizer and $d = 3$.

A. Boundary and hypothesis audit

Let $H = \ell^2$ and $M = c_{00}$, the finitely supported sequences. For $x = (1/n)_{n\geq 1}$, truncations in M approach x, so the infimum distance is zero. It is not attained because $x \notin M$. This isolates closedness.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected export (N1): every subspace admits a closest point. **Validation witness:** the c_{00} example above. **Knowledge diagnosis:** closedness was dropped when moving from a limit in H to membership in M. **Repair:** keep the stated closed-subspace hypothesis and explicitly invoke it in N1.2. This repairs the proof of the stated theorem; extending it to all subspaces is false. N2 and N3 must consume the corrected membership export.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Construct orthogonal decompositions and then the Hilbert-space representation of continuous linear functionals in FA-03.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT OpenCourseWare, Introduction to Functional Analysis: lecture notes. See the lectures on integration, Hilbert spaces, Baire category, and Hahn–Banach.